

CAMPINA GRANDE, BrazilWMO: **827950**Lat: **7.270S**Lon: **35.896W**Elev: **1646**StdP: **13.84**Time Zone: **-3.00 (W03)**Period: **99-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	64.0	64.5	56.8	73.0	84.1	57.6	75.3	83.8	19.7	77.5	18.3	77.0	4.0	130	0.354

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
3	16.6	91.2	70.5	89.7	70.8	88.2	70.6	75.0	82.1	74.3	81.4	73.7	80.8	10.7	100

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
73.5	132.8	76.8	72.7	128.8	75.7	71.9	125.5	74.9	39.6	82.2	38.8	81.3	38.3	80.3	82.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 18.8	17.6	16.3	DB	61.0	93.6	1.4	1.7	60.0	94.9	59.2	95.9	58.4	96.9	57.4	98.1	(4)
(5)			WB	60.3	78.6	1.2	2.5	59.5	80.4	58.8	81.8	58.1	83.2	57.3	85.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	76.7	78.5	78.7	79.1	78.6	76.6	74.2	72.6	73.1	74.9	76.9	78.6	78.9
	DBStd	2.98	1.81	1.86	1.88	2.26	1.93	1.98	1.83	1.84	1.96	1.78	1.50	1.84
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD50	9751	884	803	901	858	824	727	701	717	748	835	857	896
	CDD65	4276	419	383	436	408	359	277	236	252	298	370	407	431
	CDH74	26465	2913	2669	3049	2741	1994	1165	805	1127	1624	2339	2934	3105
	CDH80	8830	1093	1002	1162	1004	536	194	81	171	404	785	1167	1231
Wind	WSAvg	9.1	9.8	9.1	8.6	7.8	7.6	7.9	8.7	9.8	10.4	10.9	10.6	
Precipitation	PrecAvg	34.0	2.4	2.8	3.9	3.7	4.1	5.6	4.4	2.6	1.3	0.5	0.4	0.9
	PrecMax	59.2	15.8	9.7	10.6	9.0	13.7	11.9	10.9	7.4	5.8	2.1	2.0	2.5
	PrecMin	17.4	0.2	0.4	0.2	0.3	0.4	1.0	0.0	0.0	0.2	0.0	0.0	0.0
	PrecStd	10.8	3.0	2.6	2.2	2.2	2.7	2.7	2.3	1.7	1.4	0.4	0.5	0.7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	91.2	91.5	91.8	93.5	88.2	86.0	84.0	85.1	87.8	89.6	91.4	91.5
		MCWB	70.4	70.3	71.0	70.9	71.1	70.4	69.4	68.6	69.8	70.3	69.7	70.4
	2%	DB	89.4	89.5	89.7	91.0	86.4	84.0	82.3	83.8	85.8	87.8	89.5	89.6
		MCWB	70.8	71.2	71.4	70.9	71.5	70.4	69.2	68.4	69.2	69.6	70.2	70.5
	5%	DB	87.7	87.9	87.9	88.1	85.3	82.4	80.6	82.2	84.2	86.2	87.9	88.0
		MCWB	70.4	71.1	71.8	70.9	71.5	70.1	68.7	68.3	68.8	69.8	69.9	70.3
	10%	DB	86.1	86.1	86.2	86.0	82.8	80.6	78.8	80.4	82.5	84.4	86.3	86.3
		MCWB	70.7	71.3	72.0	71.8	71.4	70.0	69.0	68.4	68.8	69.6	70.0	70.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	74.5	75.5	75.9	76.1	75.4	74.3	72.9	72.0	72.8	74.3	73.6	74.5
		MCDB	83.0	82.5	83.3	82.1	80.8	79.3	77.7	77.8	81.3	83.5	84.3	83.8
	2%	WB	73.4	74.3	74.5	74.6	74.2	72.9	71.8	70.9	71.6	72.0	72.0	72.8
		MCDB	81.5	81.6	82.3	81.3	79.9	78.7	76.9	77.4	79.3	81.5	82.4	81.8
	5%	WB	72.7	73.4	73.8	73.8	73.4	72.1	70.9	70.1	70.8	71.1	71.3	72.2
		MCDB	80.4	80.9	81.5	80.2	79.7	77.5	75.9	77.1	78.3	80.2	81.2	80.6
	10%	WB	72.1	72.7	73.1	73.3	72.8	71.4	70.2	69.5	70.1	70.6	70.9	71.6
		MCDB	79.4	80.0	80.5	79.5	78.6	76.4	75.1	76.4	77.5	79.3	80.2	79.5
Mean Daily Temperature Range	5% DB	MDBR	16.9	16.6	16.6	16.2	14.7	13.6	13.1	15.5	16.5	17.8	18.6	17.3
		MCDBR	19.0	18.7	18.9	20.3	17.3	16.4	16.4	18.1	18.5	19.3	19.6	19.0
		MCWBR	4.6	4.2	4.3	4.6	5.3	5.6	6.0	5.6	5.0	5.0	4.0	4.0
	5% WB	MCDBR	16.3	15.8	16.0	15.2	14.0	13.2	12.5	14.7	15.9	17.5	18.3	17.1
		MCWBR	5.0	5.1	4.7	5.1	5.5	5.8	6.0	6.3	5.6	5.6	4.4	4.5
Clear-Sky Solar Irradiance	taub	0.373	0.378	0.382	0.378	0.360	0.360	0.361	0.360	0.379	0.385	0.375	0.387	
	taud	2.494	2.486	2.478	2.497	2.549	2.530	2.509	2.472	2.395	2.401	2.441	2.428	
	Ebn at Noon	306	304	298	291	287	282	284	292	294	298	303	301	
	Edh at Noon	36	37	36	34	31	31	32	35	39	40	38	39	
All-Sky Solar Radiation	RadAvg	1888	1876	1859	1716	1568	1414	1487	1680	1853	1970	2041	1948	
	RadStd	126	100	85	95	95	98	102	89	95	111	111	77	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (3 neighbors)	-0.85	-0.54	-1.17	-1.71	-0.83	-0.59	N/A	N/A	-392	-392

Nomenclature: See separate page